

ALISTRO, France

WMO: 077750

Lat: 42.2597N

Lon: 9.5414E

Elev: 243

StdP: 14.57

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	38.2	40.8	18.0	13.8	46.2	21.2	16.1	46.9	32.4	54.3	27.7	50.0	13.6	340	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.8	88.2	73.5	86.2	73.3	84.4	72.6	77.0	84.9	75.8	83.4	74.6	82.1	9.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.4	129.6	82.8	73.2	124.4	81.6	71.9	119.2	80.5	40.5	85.1	39.4	83.7	38.3	82.1	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.8	22.0	18.5	DB	35.2	91.5	3.0	3.4	33.0	93.9	31.2	95.9	29.5	97.8	27.3	100.2	(4)
(5)			WB	30.6	78.1	2.2	1.5	29.0	79.1	27.7	80.0	26.4	80.8	24.8	81.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.3	51.4	50.8	54.2	59.0	64.7	72.5	77.5	77.9	72.2	66.0	58.6	53.6	(6)	
	DBStd	10.46	4.49	4.87	4.25	3.67	3.90	4.55	3.37	3.40	3.90	4.02	4.67	4.62	(7)	
	HDD50	125	38	44	16	1	0	0	0	0	0	0	4	22	(8)	
	HDD65	1981	423	397	335	183	53	3	0	0	3	35	195	354	(9)	
	CDD50	4969	80	67	146	270	454	676	852	866	666	497	262	133	(10)	
	CDD65	1351	0	0	0	3	42	229	387	401	219	67	3	0	(11)	
	CDH74	8509	0	0	0	3	104	1251	3096	3151	842	61	1	0	(12)	
	CDH80	1998	0	0	0	0	5	249	786	849	108	1	0	0	(13)	
(14) Wind	WSAvg	8.7	9.3	9.6	9.7	8.8	8.5	7.8	7.7	7.6	8.3	8.5	9.1	9.3	(14)	
(15) Precipitation	PrecAvg	27.8	2.6	2.1	2.6	2.2	1.9	1.1	0.5	0.7	2.6	4.2	4.1	3.3	(15)	
	PrecMax	46.1	7.1	5.8	8.6	4.4	7.9	3.5	2.5	2.4	7.6	10.6	10.6	6.9	(16)	
	PrecMin	22.0	0.3	0.0	0.1	0.3	0.4	0.0	0.0	0.0	0.2	0.1	0.3	0.0	(17)	
	PrecStd	6.1	1.7	1.7	2.3	1.1	1.7	0.9	0.7	0.6	2.2	2.8	2.6	2.1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.3	65.0	68.5	72.9	80.5	89.5	90.4	92.7	86.5	78.0	71.6	67.7	(19)	
		MCWB	53.3	53.5	54.6	59.9	64.7	70.8	73.6	73.7	69.2	67.3	62.2	56.2	(20)	
	2%	DB	61.9	61.2	64.8	70.1	76.9	85.1	87.6	88.7	82.8	75.7	68.5	63.6	(21)	
		MCWB	52.6	52.3	53.9	59.3	65.2	71.2	74.1	73.7	71.0	66.8	60.9	54.8	(22)	
	5%	DB	59.3	59.3	62.4	67.9	74.2	82.7	85.8	86.6	80.7	73.7	66.4	61.5	(23)	
		MCWB	51.8	51.6	53.4	57.8	63.5	70.8	73.6	73.7	70.2	66.0	60.2	54.3	(24)	
	10%	DB	57.3	57.3	60.4	65.6	71.8	80.5	84.3	84.5	78.7	72.0	64.6	59.6	(25)	
		MCWB	50.7	50.6	52.8	56.8	62.3	69.8	72.7	73.1	68.5	65.2	58.9	53.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.6	56.6	58.5	63.2	70.0	75.5	78.4	78.3	75.7	71.5	65.4	59.4	(27)	
		MCDB	61.2	61.0	64.1	70.2	75.7	84.2	86.7	86.9	81.3	74.5	68.3	63.0	(28)	
	2%	WB	54.9	54.7	56.7	61.1	67.1	73.9	77.0	77.2	73.7	69.4	63.3	57.8	(29)	
		MCDB	59.3	58.7	61.5	67.3	73.0	81.7	85.2	85.3	79.8	72.8	66.5	61.5	(30)	
	5%	WB	53.4	53.4	55.6	59.7	65.3	72.6	75.8	76.1	72.1	68.0	61.6	56.2	(31)	
		MCDB	57.8	57.6	59.7	65.3	71.7	80.1	83.6	83.8	78.6	72.1	64.7	60.0	(32)	
	10%	WB	52.0	51.8	54.5	58.5	63.8	71.1	74.7	74.8	70.4	66.6	60.2	54.5	(33)	
		MCDB	56.4	55.9	58.5	63.7	70.3	78.6	82.2	82.3	77.0	70.8	63.6	58.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	10.2	10.6	11.2	11.6	12.2	12.0	11.8	11.1	10.2	9.5	9.5	(35)	
		MCDBR	11.9	12.4	13.7	13.9	14.0	13.9	13.3	13.9	13.0	11.9	10.9	11.3	(36)	
	5% WB	MCWBR	8.9	9.5	10.4	9.4	9.7	9.4	9.5	10.6	9.6	8.9	8.2	8.7	(37)	
		MCWBR	10.6	11.3	11.6	12.0	12.1	12.4	12.4	12.6	11.9	10.3	9.1	9.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.323	0.340	0.369	0.405	0.416	0.408	0.407	0.404	0.395	0.376	0.354	0.326	(40)	
		taud	2.524	2.467	2.401	2.304	2.316	2.378	2.382	2.400	2.418	2.466	2.473	2.522	(41)	
	Ebn at Noon		261	274	278	274	273	275	274	271	266	257	245	249	(42)	
		Edh at Noon	23	28	33	40	40	38	37	36	33	28	24	21	(43)	
(44) All-Sky Solar Radiation	RadAvg		550	833	1217	1630	1967	2228	2270	1977	1489	996	615	488	(44)	
	RadStd		52	79	125	97	145	110	92	96	69	85	56	52	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (15 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+128	N/A	

Nomenclature: See separate page